

Dynamical-Theory computation of Multiple-Beam Diffraction

(Latest update October 10, 2025)

I. Coordinate System (CS):

Fig. 1(a) shows the Cartesian CS for computing **Multiple-Beam Diffraction** (MBD), where the **Z**-axis is parallel to the diffraction vector **G** of the **Primary Reflection**, the **X**-axis is the **Horizontal Reference Direction** the user must choose, and **Y** = **Z** × **X**. The direction of the incident beam is described by the **diffraction angle** θ (relative to the Bragg plane) and the **azimuthal angle** Φ [relative to **X**-axis, $\Phi > 0$ along the arrow direction of Φ (toward +**Y**) in Fig. 1(a), and $\Phi < 0$ along the opposite direction]. So the incident wavevector is $\mathbf{K}_0 = K(\cos \Phi \cos \theta \hat{\mathbf{x}} + \sin \Phi \cos \theta \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}})$, and the diffracted wavevector of the primary reflection is $\mathbf{K}_G \approx K(\cos \Phi \cos \theta \hat{\mathbf{x}} + \sin \Phi \cos \theta \hat{\mathbf{y}} + \sin \theta \hat{\mathbf{z}})$, where $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ are unit vectors along **X**, **Y**, **Z**, respectively, and $K = 1/\lambda$ (λ the incident X-ray wavelength).

The **diffracting lattice planes**, also called the Bragg planes, are perpendicular to the **diffraction vector**. If the **crystal surface** is parallel to the Bragg plane, the configuration is called the **symmetric Bragg case** (of two-beam diffraction). In general, the crystal surface has an **offcut angle** with respect to the Bragg planes (of the primary reflection), denoted by Ω_{off} in Fig. 1(b). Meanwhile, the **offcut direction** is described by the azimuthal angle Φ_{off} . Note that the offcut can dramatically change the diffraction properties. For example, Fig. 1(c) shows the diffraction geometry for $\Phi_{\text{off}} = 180^\circ$ and $\Omega_{\text{off}} = 90^\circ$, which becomes a **symmetric Laue case** (for a finite crystal thickness).

II. Steps to Set Parameters:

To run the program, you first set the **Photon Energy**. Then import a **crystal structure** and select the **primary reflection** (so the **Z**-axis is determined), followed by choosing the **Horizontal Reference Direction** (hkl) to determine the **X**-axis, *i.e.* **X** is parallel to the *reciprocal lattice vector* $h\mathbf{a}^* + k\mathbf{b}^* + h\mathbf{c}^*$ in *reciprocal space*, NOT in real space.¹

Fig. 1(a) is based on the **two-beam diffraction** geometry (for 000 and **G**). When θ is fixed at the Bragg angle θ_B while Φ is varied, reflection **G** is always activated. However, at discrete angles Φ , the incident beam may also satisfy the Bragg conditions of other reflections in 3D space, giving rise to MBD. In the program, in addition to the **first** reflection $\mathbf{G}_0 = 000$

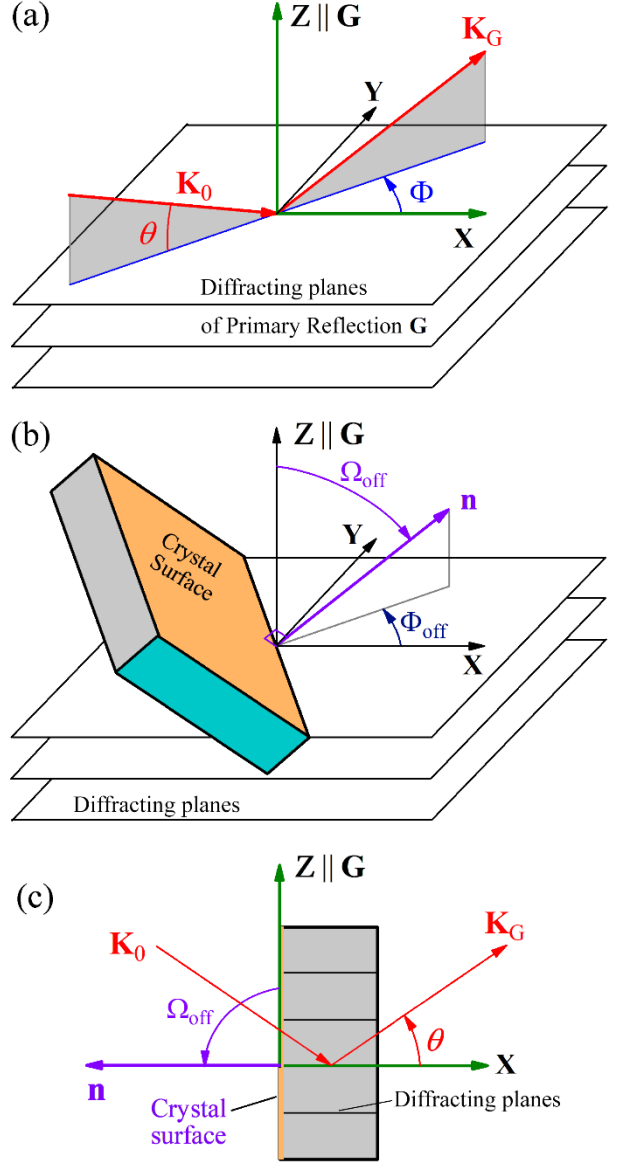


Fig. 1. (a) The CS XYZ for calculating MBD. (b) Description of the Crystal Surface in the CS by the Ω_{offcut} and Φ_{off} angles. **n** is the crystal surface normal. (c) The symmetric Laue case with $\Phi_{\text{off}} = 180^\circ$ and $\Omega_{\text{off}} = 90^\circ$.

¹ For cubic structures, $h\mathbf{a}^* + k\mathbf{b}^* + h\mathbf{c}^*$ in reciprocal space is parallel to vector $h\mathbf{a} + k\mathbf{b} + h\mathbf{c}$ in real space, but note that for non-cubic structures, these two vectors (directions) may NOT be parallel.

and the **second** reflection $\mathbf{G}_1 = \mathbf{G}$, the button **Add Reflection** allows you to add a **third reflection $\mathbf{G}_2$** (here we use **0-based indices**). Then the program calculates the Φ angles (usually two, denoted by Φ_{3BD} here) that fulfill the **three-beam diffraction** condition. You can select either angle (by clicking the **Set Azimuth Angle of 3-Beam Diffraction** or the **2nd** button) as the **Principal Azimuth angle of Incident Beam**. Around (Φ_{3BD}, θ_B) you can now calculate the 3-beam **diffraction intensity contour** in terms of (Φ, θ) of the Bragg planes --- NOT the crystal surface! In fact, you can add more reflections and you may choose any Principal Azimuth angle of Incident Beam to compute MBD. If the Bragg condition of a reflection you included is not satisfied, it would not affect the computing results! If none of the additional reflections is effective, the program gives you the conventional **two-beam diffraction** (000 and $\mathbf{G}$) results! Therefore, you can add as many reflections as you like, even irrelevant reflections!---but more reflections could slow down the computing.

You can click **Calculate or Import Chi Components** to obtain the complex **Chi (χ) components** (proportional to the **structure factors**). For N -beam diffraction, the Chi components form a $N \times N$ matrix

$$\begin{pmatrix} \chi_{0-0} & \chi_{0-1} & \cdots & \chi_{0-(N-1)} \\ \chi_{1-0} & \chi_{1-1} & \cdots & \chi_{1-(N-1)} \\ & & \vdots & \\ \chi_{(N-1)-0} & \chi_{(N-1)-1} & \cdots & \chi_{(N-1)-(N-1)} \end{pmatrix}$$

where the **0-based indices ($m - n$)** of each component indicate its corresponding diffraction vector $\mathbf{G}_m - \mathbf{G}_n$ (with $\mathbf{G}_0 = 000$, $\mathbf{G}_1 = \mathbf{G}$, $\mathbf{G}_3$, $\mathbf{G}_4$, ... $\mathbf{G}_{N-1}$ being the involved reflections). The program can automatically calculate the χ components, or you can import your own χ data from a text file.

The **polarization state** is always defined with respect to the two-beam diffraction configuration of Fig. 1(a). For **Sigma (σ) polarization**, $\mathbf{E}^\sigma = E^\sigma (-\sin \Phi \hat{\mathbf{x}} + \cos \Phi \hat{\mathbf{y}})$, and the magnetic field $\mathbf{H}$ of the **incident beam** lies in the plane containing both $\mathbf{K}_0$ and $\mathbf{G}$ and meanwhile is perpendicular to $\mathbf{K}_0$, i.e. $\mathbf{H}^\sigma = H^\sigma (\cos \Phi \sin \theta \hat{\mathbf{x}} + \sin \Phi \sin \theta \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}})$. For **Pi (π) polarization** of $\mathbf{E}^\pi = E^\pi (\cos \Phi \sin \theta \hat{\mathbf{x}} + \sin \Phi \sin \theta \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}})$, $\mathbf{H}$ is perpendicular to both $\mathbf{K}_0$ and $\mathbf{G}$, i.e., $\mathbf{H}^\pi = H^\pi (-\sin \Phi \hat{\mathbf{x}} + \cos \Phi \hat{\mathbf{y}})$. In the MBD program, the **Polarization angle** allows you to define a mixed polarization state. When the polarization angle is 0, it corresponds to pure Sigma polarization. When the polarization angle is 90°, it corresponds to pure Pi polarization. Between 0 and 90°, it is mixed polarization, and the **Polarization angle** is the angle between the **E** wave amplitude and the “**Sigma Axis**” $\mathbf{n}^\sigma$ of $\mathbf{K}_0$ [see §III Polarization Analyses later]. Currently only **linear polarization** of the incident beam is implemented. **Elliptical polarization** of incidence can be easily implemented in the future if users request. (Or you can do it yourself. First calculate the Sigma polarization results, and then separately calculate another set of data for Pi polarization. Then during your own data processing, you may add a phase factor to the output of the Pi wave amplitudes. You may also need to add a magnitude factor to the Pi results based on the strength ratio of your incident Sigma and Pi components of your elliptical incident wave. Then add the two results together.)

The parameter ΔE allows you to compute MBD by slightly deviating the **X-ray energy** from the Bragg **Photon Energy** you specified at the beginning. The $\Delta\theta$ and $\Delta\Phi$ scanning ranges allow you to calculate the 2D MBD intensity contour as a function of (Φ, θ) . If you want a 1D Darwin curve in terms of only θ (or Φ), you can simply **set the points of the other angular parameter, Φ (or θ), to be ONE (1)**. Then the other parameter is the starting value of the range and will be contestant. The “Si/Ge/C structure” choice is for the diamond structure (Si, Ge, diamond). This choice is automatically checked if the crystal name is one of “Si”, “Ge”, “C”, “Silicon”, “Germanium”, “Diamond”. You can manually check/de-check it. Checking this choice can significantly speed up the computing for Si/Ge/C crystals.

The 3D vectorial dynamical theory of MBD used in the program is based on the **Fourier coupled-wave diffraction theory (FCWDT)** by X.-R. Huang, et al., *Phys. Rev. A* **87**, 063828 (2013) (Section IV). In this theory, each involved reflection $\mathbf{G}_m$ activates a diffracted wave $\mathbf{E}_m^R \exp(-i\mathbf{K}_m^R \cdot \mathbf{r})$ above the crystal surface [with $\mathbf{E}_0^R \exp(-i\mathbf{K}_0^R \cdot \mathbf{r})$ being the specular reflection of the surface] ($m = 0, 1, 2 \cdots, N - 1$). If the crystal is a parallel plate with a finite thickness, there are

another N waves $\mathbf{E}_m^T \exp(-i\mathbf{K}_m^T \cdot \mathbf{r})$ below the exit surface [with $\mathbf{E}_0^T \exp(-i\mathbf{K}_0^T \cdot \mathbf{r})$ being the forward transmitted wave], which is true for both the Bragg and Laue cases.

For simplicity, this program does NOT provide a graphic interface for visualizing the MBD results. Instead, it always opens a file for you to export the computed results. Afterwards, you can use third-party software (e.g. ORIGIN) to plot the contour or curves using the file. Usually, you are interested in the reflectivity of the primary reflection ($\propto |\mathbf{E}_1^R|^2$ for the Bragg case or $\propto |\mathbf{E}_1^T|^2$ for the Laue case). But the program allows you to export the reflectivity of any reflection $m = 0, 1, 2, \dots, N - 1$. The default format of the output file has three columns:

$$\Delta\Phi, \Delta\theta, R_m$$

where the unit of $\Delta\Phi$ and $\Delta\theta$ can be either arcsec or μrad , and R_m is the (total) reflectivity of the m th reflection (for $m = 0, 1, 2, \dots, N - 1$). If you use Origin, you can import the three-column data. Then select the third column, C(Y), and use Menu “Column → Set As” to set this column as Z. Then use Menu “Plot → Contour/Heat Map → Color Fill” to plot the contour. By default, Origin may plot the contour with “Speed Mode is On”. You can use Menu “Graph → Speed Mode → Open Dialog” to turn off the Speed Mode.

For semi-infinite crystals (Crystal Thickness is -1), you can calculate the reflectivity only on the Entrance Crystal Surface. For finite-thickness crystals, you may calculate the transmission reflectivity on the Exit Crystal Surface Side. For the Laue transmission case of a beam [such as Fig. 1(c)], only the transmission reflectivity has noticeable values in general.

III. Polarization Analyses:

The MBD program also allows you to export the amplitude components and the corresponding reflectivity components along two orthogonal directions perpendicular to the wavevector of the m th reflection. This option is mainly for designing X-ray phase plates (to control X-ray polarization) using two-beam Bragg diffraction. So here as an example, we discuss two-beam diffraction in Fig. 1(a) (for $\Phi = 0$). Note that for a plane wave, its $\mathbf{E}$ and $\mathbf{H}$ fields are always perpendicular to the wavevector. For the two-beam diffraction in Fig. 1(a), we define the Sigma Axis as $\mathbf{n}^\sigma = \mathbf{K}_0 \times \mathbf{K}_G / |\mathbf{K}_0 \times \mathbf{K}_G|$. The Sigma Axis is the same for both $\mathbf{K}_0$ and $\mathbf{K}_G$. The Pi Axis of $\mathbf{K}_0$ is defined as $\mathbf{n}^\pi = \mathbf{K}_0 \times \mathbf{n}^\sigma / |\mathbf{K}_0|$, and the Pi Axis of $\mathbf{K}_G$ is defined as $\mathbf{n}_G^\pi = \mathbf{K}_G \times \mathbf{n}^\sigma / |\mathbf{K}_G|$. Here in general $\mathbf{n}_G^\pi \neq \mathbf{n}^\pi$. Note that $\mathbf{n}^\sigma$, $\mathbf{n}^\pi$, $\mathbf{K}_0$ are perpendicular to each other. $\mathbf{n}^\sigma$, $\mathbf{n}_G^\pi$, and $\mathbf{K}_G$ are also perpendicular to each other. In the above, we have mentioned that the $\mathbf{E}_{\text{in}}$ wave amplitude of the incident wave can be inclined by a Polarization angle φ_0 from $\mathbf{n}^\sigma$ toward $\mathbf{n}^\pi$, i.e. $\mathbf{E}_{\text{in}} = E_{\text{in}}(\cos \varphi_0 \mathbf{n}^\sigma + \sin \varphi_0 \mathbf{n}^\pi)$.

Now we define another two orthogonal unit vectors $\hat{\alpha}$ and $\hat{\beta}$ that are rotated from $\mathbf{n}^\sigma$ and $\mathbf{n}^\pi$, respectively by an angle φ_0 around $\mathbf{K}_0$,

$$\hat{\alpha} = \cos \varphi_0 \mathbf{n}^\sigma + \sin \varphi_0 \mathbf{n}^\pi, \quad \hat{\beta} = -\sin \varphi_0 \mathbf{n}^\sigma + \cos \varphi_0 \mathbf{n}^\pi,$$

so that $\hat{\alpha}$ is parallel to $\mathbf{E}_{\text{in}}$ (i.e., $\mathbf{E}_{\text{in}} = E_{\text{in}}\hat{\alpha}$) and is the incidence polarization direction. $\hat{\beta}$ is perpendicular to the incidence polarization direction. Note that $\hat{\alpha}$ and $\hat{\beta}$ are the same for the incident wave and the forward transmitted wave $\mathbf{E}_0^T \exp(-i\mathbf{K}_0^T \cdot \mathbf{r})$. Similarly, for the diffracted wave we define two orthogonal unit vectors $\hat{\alpha}_G$ and $\hat{\beta}_G$ that are rotated from $\mathbf{n}^\sigma$ and $\mathbf{n}_G^\pi$, respectively by an angle φ_G around $\mathbf{K}_G$,

$$\hat{\alpha}_G = \cos \varphi_G \mathbf{n}^\sigma + \sin \varphi_G \mathbf{n}_G^\pi, \quad \hat{\beta}_G = -\sin \varphi_G \mathbf{n}^\sigma + \cos \varphi_G \mathbf{n}_G^\pi.$$

$\hat{\alpha}_G$ and $\hat{\beta}_G$ are perpendicular to $\mathbf{K}_G$. The default value of φ_G is the same as φ_P (i.e. by default $\varphi_G = \varphi_P$).

Based on these setups, you can choose the complete format of the output file with the following **columns**:

$\Delta\Phi$, $\Delta\theta$, R_m , R_m^α , R_m^β , $|H_m^{R,\alpha}|$, phase of $H_m^{R,\alpha}$ (deg.), $|H_m^{R,\beta}|$, phase of $H_m^{R,\beta}$ (deg.), K_{mx} , K_{my} , K_{mz}

for **Reflection (Entrance Crystal Surface side)**, or

$\Delta\Phi$, $\Delta\theta$, T_m , T_m^α , T_m^β , $|H_m^{T,\alpha}|$, phase of $H_m^{T,\alpha}$ (deg.), $|H_m^{T,\beta}|$, phase of $H_m^{T,\beta}$ (deg.), K_{mx} , K_{my} , K_{mz}

for **Transmission (Exit Crystal Surface side)** (only available if the crystal thickness is NOT -1), where $m = 0, 1$, and $m = 1$ corresponding to the primary reflection $\mathbf{G}$. You do not need to export all the columns; you can select which columns you want for export. But you must remember the meaning of each column in your output file.

In the above columns, $\Delta\Phi$ and $\Delta\theta$ are the relative incidence angles (you can ignore $\Delta\Phi$ for two-beam diffraction). R_m (or T_m) is the **total reflectivity**. R_m^α and R_m^β are the two **reflectivity components** along $\hat{\alpha}$ and $\hat{\beta}$ (or $\hat{\alpha}_G$ and $\hat{\beta}_G$), and $H_m^{R,\alpha}$ and $H_m^{R,\beta}$ are the corresponding (complex) wave amplitudes (exported as the modulus and phase). **Note the output wave amplitudes are for the magnetic field $\mathbf{H}$, not for the electric field $\mathbf{E}$. $\mathbf{E}$ is perpendicular to $\mathbf{H}$ and the corresponding wavevector.**

Defining $\hat{\alpha}$, $\hat{\beta}$, $\hat{\alpha}_G$ and $\hat{\beta}_G$ are mainly for studying polarization changes for phase plates. The incident polarization is always along $\hat{\alpha}$ with zero $\hat{\beta}$ component.

For synchrotron radiation beams, in general $\hat{\alpha}$ is the horizontal direction and $\hat{\beta}$ is the vertical direction at the beamline for phase plates.

For **Phase Plates**, you may follow the following steps: ---

1. Launch the **Multiple-Beam Diffraction** (MBD) program, Set the **Photon Energy** you want, and Set the **Crystal Thickness** (you can change it any time later).
2. Click “Define Crystal” to select a crystal structure and choose the **primary reflection** (say 111).
3. We are using two-beam diffraction, but we still need to set the “**Horizontal Reference Direction**” (say 1 -1 0) although it does not affect the calculation.
4. Set the “**Principal Azimuth angle of Incident Beam**” to 0.
5. Don’t Add any addition reflection! Just click “**Calculate or Import Chi Components**” to “**Automatically Calculate**” the chi values.

6. Set the “**Polarization angle**”.

The screenshot shows the "Dynamical Theory Computation of Multiple-Beam Diffraction" window. The interface includes several input fields and controls:

- Photon Energy...:** Set to 15 KeV.
- Crystal Thickness:** Set to 50 μm.
- Si//Ge/C structure:** Checked.
- Define Crystal ...:** Set to Silicon.
- Primary Reflection (G):** Set to 1 1 1.
- $\theta_B =$** Set to 7.5737783 °.
- Horizontal Reference Direction...:** Set to 1 -1 0, corresponding to Azimuthal angle = 0.
- Crystal Surface Offcut angle from Bragg plane (0 - 180):** Set to 0 °.
- Azimuthal angle of Offcut :** Set to 0 °.
- Principal Azimuth angle of Incident Beam:** Set to 0 °.
- Set Azimuth Angle of 3-Beam Diffraction:** Set to 2nd based on 000, G and Reflection #2 in the list below.
- Add Reflection:** Three spinners set to 1, 1, 1.
- Remove selected reflection:** Button.
- Table:** A table with columns: Reflection #, h k l, Bragg angle (deg), and Azimuthal angles (deg).
- Polarization angle:** Set to 45 °.
- From 0 - Sigma To 90 - Pi:** Label.
- Calculate or Import Chi Components:** Button.
- $\Delta E =$** Set to 0 eV.
- $\Delta\theta$ from:** Set to -20 arcsec.
- to:** Set to 30 arcsec.
- arcsec:** Label.
- points:** Set to 50 points.
- $\Delta\Phi$ from:** Set to 0 arcsec.
- to:** Set to 1 arcsec.
- arcsec:** Label.
- points:** Set to 1 point.
- Calculate Multiple-Beam Diffraction with Plane-Wave Incidence:** Button.
- Divergent/polychromatic Incidence Beam:** Section header.
- FWHM:** Input field.
- eV:** Unit label.
- Integration Range:** Input field.
- eV:** Unit label.
- points:** Input field.
- Beam Divergence:** Input field.
- μrad:** Unit label.
- points:** Input field.
- Deactivate:** Button.
- Activate Convolution Panel:** Button.
- Search Reflections with Azimuth Angles close to current Principal Azimuth Angle of Incidence:** Section header.
- $|\Delta\Phi| <=$** Set to 5 °.
- h k l:** Column header for a table.
- Delta Phi (deg):** Column header for a table.
- 1e-08:** Value.
- Multiple-Beam Diffraction Lines (Monochromator Glitches) in Azimuth-Energy Map...:** Section header.

- Now click “**Calculate Multiple-Beam Diffraction with Plane-Wave Incidence**” to get the following “**Save MBD results to a text file**” dialog.

Save MBD results to a text file

For simplicity, this program did not provide an interface for plotting the MBD results. Instead, it saves the results in a text file for you to use third-party software (e.g. Origin) to plot the intensity contours or curves.

Save Results as: C:\MyXRD\al.dat

Select a diffracted beam. 0-based Beam #
(0 - Specular Reflection or Forward Transmission,
1 - Primary Reflection)

0

☐ Reflection (Entrance Crystal Surface side)
☒ Transmission (Exit Crystal Surface Side)

Columns of Data in the Exported File

☐ Delta Phi ☒ Delta theta ☒ Reflectivity or Intensity (a.u.) ☒ Reflectivity components ☐ Wave Amplitudes ☐ Diffracted Wavevector

☒ Arcsec ☐ urad

45 deg. - Rotation from s and p axes

During the following lengthy computing, which could be extremely time-consuming, please Don't close the program unless absolutely necessary!!!

Cancel Let's GO!!!

8. Choose the file name “Save Results as”.
9. Select “Transmission” and select a diffracted beam as “0”, which will select the **forward transmitted beam**.
10. De-select “Delta Phi”, and select “Reflectivity components” and leave the Rotation angle default, which is always the same as the Polarization angle you set before.
11. Click “Let’s GO!!!” and you will get a file with columns “Delta-Theta”, “Total transmissivity” and two transmissivity components. The first component is along the original incidence polarization direction, and the second component is along the direction perpendicular to the original polarization direction.
12. You can change any parameters and repeat the above processes.
13. You can also select “Reflection” and select beam 1 in the above “Save MBD results to a text file” dialog to calculate the diffraction Reflectivity and components (for the Bragg reflection geometry).
14. If you want to calculate 2D Maps, you can set a range for “Delta Phi” and select “Delta Phi” for the output.

The above procedures work for any wave in N -beam diffraction $N \geq 2$. It allows you to check the polarization of each wave along any two orthogonal directions ($\hat{\mathbf{a}}_m^W$ and $\hat{\mathbf{b}}_m^W$) perpendicular to the wavevector. For any wave $\mathbf{E}_m^W \exp(-i\mathbf{K}_m^W \cdot \mathbf{r})$ with $m > 0$ and $W = R$ or T , its **Sigma Axis** is $\mathbf{n}_m^{W,\sigma} = \mathbf{K}_0 \times \mathbf{K}_m^W / |\mathbf{K}_0 \times \mathbf{K}_m^W|$, and its **Pi Axis** is $\mathbf{n}_m^{W,\pi} = \mathbf{K}_m^W \times \mathbf{n}_m^{W,\sigma} / |\mathbf{K}_m^W|$. And you can define any rotation angle φ_m^W for this wave to set the two orthogonal axes

$$\hat{\mathbf{a}}_m^W = \cos \varphi_m^W \mathbf{n}_m^{W,\sigma} + \sin \varphi_m^W \mathbf{n}_m^{W,\pi}, \quad \hat{\mathbf{b}}_m^W = -\sin \varphi_m^W \mathbf{n}_m^{W,\sigma} + \cos \varphi_m^W \mathbf{n}_m^{W,\pi}.$$

An exception is that the **Sigma Axis** of the incident wavevector $\mathbf{K}_0$ is always defined as $\mathbf{n}^\sigma = \mathbf{K}_0 \times \mathbf{K}_G / |\mathbf{K}_0 \times \mathbf{K}_G|$, which is related to the primary reflection wavevector $\mathbf{K}_G$. The **Pi Axis** of $\mathbf{K}_0$ is always $\mathbf{n}^\pi = \mathbf{K}_0 \times \mathbf{n}^\sigma / |\mathbf{K}_0|$, which is also dependent on $\mathbf{K}_G$. $\mathbf{n}^\sigma$ and $\mathbf{n}^\pi$ are also the Sigma and Pi Axes of the forward transmitted wave. The two axes of the specular reflection wave $\mathbf{E}_0^R \exp(-i\mathbf{K}_0^R \cdot \mathbf{r})$ is $\mathbf{n}_0^{R,\sigma} = \mathbf{K}_0 \times \mathbf{K}_0^R / |\mathbf{K}_0 \times \mathbf{K}_0^R|$ and $\mathbf{n}_0^{R,\pi} = \mathbf{K}_0^R \times \mathbf{n}_0^{R,\sigma} / |\mathbf{K}_0^R|$.

IV. Divergent/polychromatic Incident Beam

The above computation is based on a plane-wave incident beam, where you calculated the reflectivity as a function of $(\Delta\Phi, \Delta\theta)$. At the bottom, by clicking “**Activate Convolution Panel**”, the program also allows you to use an incident beam with divergence along both the Φ and θ directions and a finite (FWHM) spectral bandwidth (Gaussian shape). Then for each point $(\Delta\Phi, \Delta\theta)$ --- the same parameters in the above Plane-Wave panel --- the program makes multiple integration over the two **Beam Divergence Ranges** and the **energy Integration Range**. Therefore, the multiple integration could be extremely slow, and may take hours to days. For testing the convergence, you may limit the integral over small ranges and make a **1D scan of $\Delta\Phi$ or $\Delta\theta$** --- don’t do the 2D $(\Delta\Phi, \Delta\theta)$ mapping during testing. Then you can test each parameter of the Range or Points in the Convolution Panel. When the results show no noticeable difference, you reach the convergence for this parameter. After you think all the parameters reach the convergence requirements, you may make full scan of **$\Delta\Phi$ and/or $\Delta\theta$** .

V. Reflections with Azimuth Angles (Φ) close to the Principal Azimuth Angle of Incidence

The right panel of the MBD program is “**Check Reflections with Azimuth Angles close to current Principal Azimuth Angle of Incidence**”. It searches, in the vicinity of the current **Principal Azimuth Angle** (you must set), if there are other reflections with their Bragg conditions nearly satisfied. For example, at $E = 8$ keV, Primary Reflection Si 004, Horizontal Reflection Direction 100. The Φ angle of Reflection 111 is -38° . If you set the Principal Azimuth Angle to be -38° , you would think that now it is a 000-004-111 three-beam diffraction condition. However, if you check other reflections in the right panel, you can find that reflection 113 also has the same Φ angle ($\Delta\Phi = 0$). Therefore, this is, in fact, a **000-004-111-113 four-beam diffraction** configuration.

You may include the searched reflections in the MBD computing in the list of the left panel if there $\Delta\Phi$ values are very close to zero. Note that some of the searched reflections may be **forbidden reflections**. However, a three-beam diffraction ($000-\mathbf{G}_1-\mathbf{G}_2$) process actually involves a hidden diffraction channel $\mathbf{G}_{\text{hidden}} = \mathbf{G}_1 - \mathbf{G}_2$. The three-beam diffraction process still occurs if only two of the three reflections $\mathbf{G}_{\text{hidden}}$, $\mathbf{G}_1$, $\mathbf{G}_2$ are allowed reflections. For example, Si 000-111-002 three-beam diffraction exists because it involves reflections 111, 002, and -1-11. Reflections 111 and -1-11 are allowed reflections although 002 direct reflection is forbidden, i.e., the diffraction process is a detour reaching 002 through 111 and -1-11.

The button “**Multiple-Beam Diffraction Lines (Monochromator Glitches) in Azimuth-Energy Map...**” allows you to activate the “**Multiple-Beam Diffraction Lines (Monochromator Glitches) in Azimuth-Energy Map**” that can provide a full Φ - E map of the MBD lines *based on the current crystal and configuration*, see the following section.

VI. MBD lines in Azimuth-Energy ($\Phi - E$) Coordinate System

This program provides fast mapping of **multiple-beam diffraction lines**, also called **Monochromator Glitches** in the EXAFS community, in the Azimuth-Energy space. It particularly includes those **forbidden reflections with allowed detour reflections**, such as the above Si 000-111-002 three-beam diffraction case, i.e., for a three-beam diffraction case $000-\mathbf{G}_1-\mathbf{G}_2$, if two of the three involved reflections $\mathbf{G}_1$, $\mathbf{G}_2$, $\mathbf{G}_{\text{hidden}} = \mathbf{G}_1 - \mathbf{G}_2$ are allowed reflections, the three-beam diffraction conditions are fulfilled [see X.-R. Huang *et al.*, [Appl. Phys. Lett. 105, 181903 \(2014\)](#); [J. Appl. Cryst. 47, 1716-1721 \(2014\)](#)]. For Si/Ge/C structures, the program uses the diamond structure rule to judge if a reflection hkl is allowed (h, k, l are all even numbers with $h + k + l = 4n$, or h, k, l are all odd numbers) or forbidden. For general crystals, the program calculates the **χ factor** of each reflection, and if its absolute value is less than a **threshold value** (default is 10^{-8}), the reflection is treated as a forbidden reflection. Note that in general, $\chi \sim 10^{-6} - 10^{-5}$ for allowed reflections. The computing algorithm has been highly optimized so that the computing is very fast for any crystals. **You can change the threshold value, or you can set it to zero if you want to ignore this restriction.**

The menu “**File → Export Map Data...**” function allows you to save the data points of the **Glitch lines** to a file so that you can plot the map using other software (e.g. ORIGIN). menu “**File → Save Points on a Vertical Line...**” allows

you to save the **Glitch points** on any vertical line $\Phi = \text{constant}$. This is useful for selecting lateral directions to avoid or optimize glitches during X-ray crystal monochromator designing.

You can click an MBD line to see its reflection index. For Si/Ge/C structure, different reflections may have identical lines overlapped together. So, if you click a line more than once, you may get different reflection indices. You can also allow the program to print the reflection index of each line. The index will be plotted at random positions on the line. So you can let the program print/hide the indices for more times to randomly change the label positions so that you can clearly identify each line when the lines or labels are overlapped.

VII. Parasitic Reflections of Back Reflections from Cubic Structures and the Continuous MBD planes

The menu command “**Tools → Parasitic Reflections of Back Reflection**” is for finding the parasitic reflections hkl of a back reflection HKL ($\theta_B \rightarrow 90^\circ$) from a cubic structure. The CMBDPs (Continuous Multiple-Beam Diffraction Planes) and HIDs [Horizontal Incidence Directions, i.e., the direction of the projection of the incident beam on the (HKL) planes] are for you to avoid Multiple-Beam Diffraction by avoiding these directions. Note that only 111 and 220 reflections of cubic diamond have no parasitic reflections.

Please refer to [X.-R. Huang et al., J. Appl. Cryst. 47, 1716 \(2014\), doi:10.1107/S160057671401930X](#).